

Solving Quadratics by Factoring

I. Trinomials

$$ax^2 + bx + c = 0$$

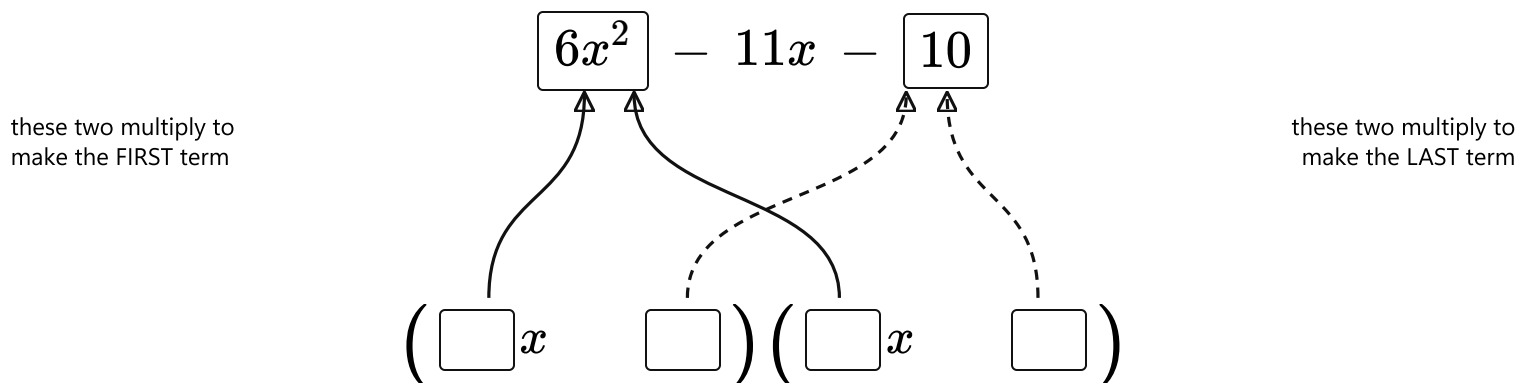
Solve the following by factoring. $12x^2 - 22x - 20 = 0$

1. Start by looking for a GCF.

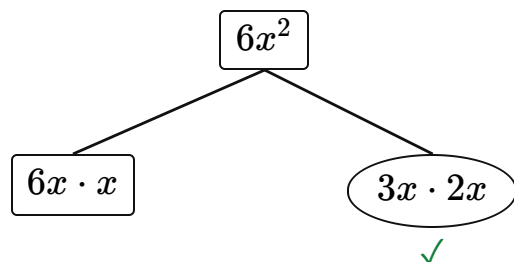
$$\xrightarrow{\text{divide by 2}} 6x^2 - 11x - 10 = 0$$

Every term divides by 2, and the equation equals zero, so divide both sides. The 2 is gone for good.

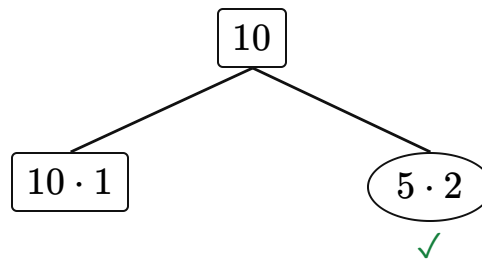
2. A trinomial factors into two sets of parentheses.



3. How can we make $6x^2$?



4. How can we make 10?



Always try the closest pair first.

5. Put the $3x$ and the $2x$ in the first spots.

It does not matter which order you put these two in.

$$\left(3x \quad \boxed{} \right) \left(2x \quad \boxed{} \right)$$

6. Now place the 5 and the 2.

Here the order does matter.

Option A

$$\left(3x \quad \boxed{5} \right) \left(2x \quad \boxed{2} \right)$$

Option B

$$\left(3x \quad \boxed{2} \right) \left(2x \quad \boxed{5} \right)$$